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Introduction to a new OCP FCSA Sub-Project Workstream
Extending OCP FCSA for Real-Time, Safety-Critical Physical AI

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What Makes Physical AI Different

A Physical AI SoC must perceive, reason, and act safely in the physical world

Its decisions directly affect physical behavior, which drives six physical AI constraints

Perceive and react

Sensor ingress
One view of the world from cameras, radars, and lidars

Deterministic timing
Bounded, repeatable latency from sensing to action

Act safely

Reliability and safety
Detect faults and reach a safe state in time

Security and trust
No untrusted actor can command physical motion



Platform-wide constraints on both

- **Power and thermal efficiency —** sustain performance in a fixed envelope
- **Multi-chiplet scalability —** one architecture across product tiers

Why Chiplets Matter for Physical AI

Chiplets enable scalable Physical AI systems, but introduce architecture challenges

Why Chiplets

- **Scalable composition**
- **Function partitioning**
- **Independent sourcing**
- **Reuse across tiers**

New challenges

- **Latency and jitter uncertainty** — no longer fixed by a single design
- **Distributed trust boundaries** — the perimeter spans chiplets and suppliers
- **Cross-chiplet fault propagation** — one die can disturb another die's control path
- **Lifecycle and configuration complexity** — discovery, update, and configuration go package-wide
- **Observability and evidence gaps** — no die can evidence the whole package

Each of these crosses a chiplet boundary, so no single supplier can resolve them alone, which makes them a standardization concern

A Physical AI profile for OCP FCSA

OCP FCSA as the baseline
The chiplet system architecture context that Physical AI builds on is already defined here.

Physical AI profile extends it
Real-time and safety-critical concerns go beyond the current baseline scope.

Behavior that crosses chiplets
OCP FCSA is the right place to standardize externally visible chiplet behavior.

There is a concrete starting point
The profile translates these constraints into ten candidate architecture requirements.

How will we classify each requirement

- Baseline OCP FCSA reuse
- Physical AI profile (required)
- Optional profile — supplier may adopt

A Physical AI profile retains existing OCP FCSA rules and adds requirements specific to real-time, safety-critical physical systems

Representative Physical AI System

All the systems are performing the same functions but differ in their processing requirements.



Autonomous vehicle

Highway and urban driving at vehicle speed



Mobile robot

Indoor and last-mile navigation among people



Industrial robot

Fixed-cell manipulation with hard real-time control



Drone

Airborne perception under tight power limits



Autonomous machine

Off-road and heavy machinery in harsh conditions

Same PAI constraints across all five, with different timing, power, and safety budgets

The Six Physical AI Constraints

Because a Physical AI SoC acts in the physical world, six constraints apply to every design.

Each constraint is what the platform must guarantee, and each one drives the architecture requirements

Sense and decide in time

Deterministic timing

Bounded, repeatable latency from sensing to action, worst case, and not just average.

Sensor ingress

A single, time-consistent view of the environment fused from camera, radar, and lidar.

Act safely and stay trusted

Reliability and safety

Detect faults, report them, and reach a safe state within the deadline.

Security and trust

No untrusted actor can command physical motion or alter safety behavior.

Sustain it across the platform

Power and thermal efficiency

Sustain performance and thermal limits in a fixed power envelope.

Multi-chiplet scalability

One architecture that grows across product tiers and suppliers.

These six constraints are the starting point of the profile

Constraints to Requirement Architecture Areas

The six physical AI constraints (C1-C6, in bold) and ten architecture requirement areas (R1-R10) that they drive

C1. Deterministic timing

R1. Closed-loop Control

R2. Deterministic Execution

R3. Real-Time Interconnect

C2. Sensor ingress

R9. Timestamped sensor data and fusion

C3. Reliability and safety

R7. Functional Safety

R8. Safety Management and Observability

C4. Security and trust

R4. Security for Physical Action

C5. Power and thermal efficiency

R5. Energy Efficiency and Power Management

R6. Heterogeneous Compute

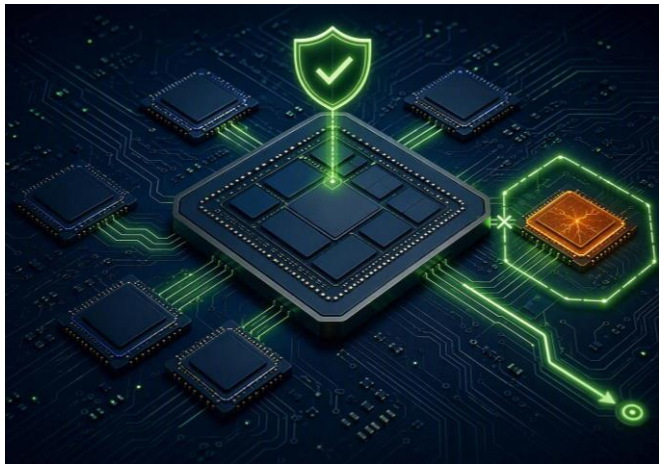
C6. Multi-chiplet scalability

R10. Multi-chiplet Scalability

Every candidate specification requirement traces back to a Physical AI constraint

Mapping Example - Functional Safety

Failures can arise in sensor interfaces, compute chiplets, die-to-die links, memory, safety controllers, or actuator paths. **Because they reach physical behavior, faults must be detected, isolated, and recovered within bounded time.**



Functional safety is a package-level property, not a per-chiplet one

The platform must deliver fault detection and containment, deterministic safety signaling, coordinated safe-state control, and degraded-mode operation.

Platform safety control and recovery

Package-level fault reporting, quarantine, safe-state entry, and coordinated recovery across chiplets.

Telemetry, debug, and trace control

Package-level fault, health, timing, and safety-state telemetry to localize faults.

Reliability and link-health reporting

Degradation and link-health trends that trigger fallback, escalation, or safe-state entry.

Traffic classes and QoS semantics

Bounded-latency handling of safety events, watchdogs, emergency stop, and safe-state commands.

Safety-critical data protection

Integrity protection for safety events, control paths, and safe-state commands across chiplets.

Verification and safety-case evidence

Evidence for fault detection, safety-event latency, recovery timing, and safe-state transition.

Six specification mechanisms cover functional safety across the package

Foundational Safety link to Physical AI



- Physical AI distributes a safety-relevant function across sensing, compute, AI inference, and actuation.
- Each chiplet becomes a potential fault domain; the D2D interconnect becomes both a fault-propagation path and a diagnostic observation point.
- Local safety mechanisms are therefore insufficient: safety must be composed across chiplets, interfaces, and system-level supervision.
- Foundational Safety provides the architectural basis for bounded, diagnosable, and controllable behavior, independent of the specific AI model.

PAI WG Scope and Charter

IN SCOPE

- Externally visible chiplet behavior required by the PAI constraints
- Classification of requirements into baseline and optional profile
- Use OCP FCSA and extend. Refer to other standards where needed.
- Compliance profiles and the evidence needed to support safety claims

OUT OF SCOPE

- Chiplet microarchitecture and implementation choices
- Die-to-die physical and link layers already owned by UCIe/BOW
- AI model, algorithm, and training behavior
- Product certification and vendor-specific software stacks

Call to Action

- Join the PAI discussion
  - Provide feedback on proposed scope & identified constraints / requirement categories
- Get involved
  - Join the mailing list & calls
- Timeline
  - Targeting specification v1.0.0 in Q3 2027
- Where to find additional information (URL links)
  - https://www.opencompute.org/wiki/Server/OCE/FCSA

Thank You